

224×224 crop

pretrained backbones

VGG16 + CBAM

5 conv blocks (ImageNet)
channel + spatial attention
global avg. pool, FC 256

ViT-B/16

16×16 patches, 12 layers
(ImageNet-21k)
[CLS] pooler output

256

768

fusion and refinement

Capsule fusion

dynamic routing ×2

256

Trans-former

4 tokens × 64
mean pool

64

heads and losses

Embedding

256-d, unit length
ArcFace s=32, m=0.3

Shape

softmax, CE ×0.2

Colour

sigmoid, BCE ×0.2